

# Muhammed Yousuf S

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## Education

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| 2023-Present | Ph.D. in Environmental Engineering<br>Civil and Environmental Engineering<br>Indian Institute of Technology Delhi | CGPA: 9.25/10 |
| 2020-22      | Master of Technology<br>Civil Engineering<br>APJ Abdul Kalam Technological University                             | CGPA: 9.49/10 |
| 2015-19      | Bachelor of Technology<br>Civil Engineering<br>APJ Abdul Kalam Technological University                           | CGPA: 7.53/10 |

## Journal Publications

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- **Yousuf, S. M.**, Karigowda, Habib, G., Warsi, A. B., Imran, M., & Nabi, M. un. (2026). Chemical investigation of polycyclic aromatic hydrocarbon sources and associated health risks in PM<sub>2.5</sub> from Eastern India. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-41899-0>
  - Shibu, K., Drisiya, J., & **Yousuf, S. M.** (2025). Spatiotemporal Modelling Approach for Nutrient Export in Sasthamkotta Freshwater Wetland Watershed. *Journal of the Indian Society of Remote Sensing*, 53(1), 225–239. <https://doi.org/10.1007/s12524-024-01978-z>
  - Shibu, K., Ayoob, S., & **Yousuf, S. M.** (2023). Determination of land use–land cover changes in the adjoining panchayats of Sasthamkotta freshwater wetland, a Ramsar site in South India. *International Journal of Lakes and Rivers*, 16(1), 17–33.
  - Deepak, K. R., Sabu, P., **Yousuf, S. M.**, & Thottumkal, C. G. (2022). Urban flood susceptibility map of Chennai using GIS and random forest method. *International Research Journal of Engineering and Technology*, 9(10).

## Technical Skills

### *Instrumentation*

- Thermal Desorption Gas Chromatography-Mass Spectrometry (TD-GC/MS)
- Organic Carbon/Elemental Carbon Analyzer (OC/EC Analyzer)
- Ion Chromatography (IC)
- Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES)
- Aethalometer
- Nephelometer
- Scanning Mobility Particle Sizer (SMPS)
- Monitor for Aerosols and Gases in Ambient Air (MARGA)
- Ambient Metals Monitor (Xact)
- Micro-Orifice Uniform Deposit Impactor (MOUDI)

### *Software and Analytical Tools*

- ArcGIS
- QGIS
- MS-Office
- Python
- Origin

## Research Presentations

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**Yousuf, S. M.,** Habib, G. Seasonal Variation, Ring Distribution, and Source Characteristics of PM<sub>2.5</sub>-bound PAHs in Eastern India. [Platform Presentation](#) at Asian Aerosol Conference 1-4 December 2025

## Workshops and Trainings

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- **Advanced Aerosol Science Intensive Course** ([View Certificate](#))  
Centre for Atmospheric Sciences (CAS), IIT Delhi and Institute for Atmospheric & Earth System Research (INAR), University of Helsinki, 19<sup>th</sup> – 29<sup>th</sup> November 2025
- **Minds-on Air quality, humaN health & climAte Nexus** ([View Certificate](#))  
Central University of Jammu & Sindhu Central University, 26<sup>th</sup> February – 1<sup>st</sup> March 2024
- **Aerosol Science Intensive Course** ([View Certificate](#))  
Centre for Atmospheric Sciences (CAS), IIT Delhi and Institute for Atmospheric & Earth System Research (INAR), University of Helsinki, 5<sup>th</sup> - 15<sup>th</sup> February 2024
- **Workshop on source apportionment** ([View Certificate](#))  
Paul Scherrer Institute (PSI) and the partners of the Clean Air Project India – Indian Institute of Technology Bombay, 15<sup>th</sup> & 16<sup>th</sup> December 2023.
- **Workshop on WebGIS Enabled Spatial Information System** ([View Certificate](#))  
TPLC, Govt. Engineering College Barton Hill Trivandrum - Directorate of Technical Education, Government of Kerala, 8<sup>th</sup> – 18<sup>th</sup> February 2021.

## Awards and Achievements

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- **Graduate Aptitude Test in Engineering 2019**  
Civil Engineering (Score: 321)
- **Civil Prize 2019**  
National Council for Technology and Training
- **Graduate Aptitude Test in Engineering 2022**  
Geomatics (Score: 450)

## Research Experience

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| <b>2023 - Pres</b> Investigating the Contribution of Organics in Refining Source Apportionment (PhD Thesis)<br>Advisor: Prof. Gazala Habib | <ul style="list-style-type: none"><li>• Investigated the chemical composition of ambient particulate matter through integrated analysis of organic, ionic, elemental, and carbonaceous (OC/EC) species.</li><li>• Developed TD–GC–MS methods for quantification of 16 EPA PAHs, alkanes, and phthalates.</li><li>• Applied Positive Matrix Factorization (PMF) for source apportionment and performed data interpretation.</li><li>• Conducted ambient aerosol measurements using both online and offline analytical techniques, integrating</li></ul> |
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| <b>2025</b>      | <p>Influence of Black Carbon Mixing State on Mass Absorption Cross-Section in the Indo-Gangetic Plain.<br/> <i>Supervisors:</i> Prof. Gazala Habib (IIT Delhi, India) and Prof. Griša Močnik (University of Nova Gorica, Slovenia)</p> | <p>real-time observations with laboratory-based chemical characterization.</p> <ul style="list-style-type: none"> <li>Investigated the physical and optical characteristics of ambient aerosols using SMPS, nephelometer, and black carbon measurements.</li> <li>Examined carbonaceous aerosol composition through integration of offline filter sampling and OC/EC analysis with real-time observations.</li> <li>Analysed PTAAM datasets alongside multi-instrument measurements for comprehensive characterization of particulate matter.</li> <li>Performed systematic data processing, quality control, and data interpretation to understand aerosol variability and atmospheric processes.</li> </ul> |
| <b>2020-2022</b> | <p>Modelling of Water Quality of Sasthamkotta Lake Using GIS and Remote Sensing (Mtech Thesis)<br/> Advisor: Prof. Shibu Krishnan</p>                                                                                                  | <ul style="list-style-type: none"> <li>Assessed spatial and temporal variations in water quality of Sasthamkotta Lake using geospatial techniques.</li> <li>Applied GIS and remote sensing tools for water quality modelling and mapping.</li> <li>Integrated satellite data and field-based observations for environmental analysis.</li> <li>Developed thematic maps to identify pollution patterns and critical zones.</li> <li>Performed data processing, analysis, and interpretation for environmental monitoring.</li> </ul>                                                                                                                                                                           |

## Positions of Responsibility

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| <b>2025</b> | <p><b>Teaching Assistantship</b><br/>(IIT Delhi)</p> | <ul style="list-style-type: none"> <li>Advanced Air Pollution Laboratory (CVP 820, Graduate course)</li> <li>Contributed to course content development and lecture preparation</li> <li>Assisted in class delivery and student engagement</li> <li>Supported laboratory experiments and practical sessions</li> <li>Assisted in preparing and grading assignments and examinations</li> </ul> |
| <b>2025</b> | <p><b>Teaching Assistantship</b><br/>(IIT Delhi)</p> | <ul style="list-style-type: none"> <li>Air Pollution and Control (CVL 7705, Graduate course)</li> <li>Assisted in course delivery and classroom instruction</li> <li>Conducted lecture sessions and facilitated student discussions</li> </ul>                                                                                                                                                |

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|             |                                               | <ul style="list-style-type: none"> <li>• Prepared and graded assignments and examinations</li> <li>• Coordinated and organized a student field visit, managing logistics and participant arrangements.</li> </ul>                                                                                                                                                                                       |
| <b>2024</b> | <b>Teaching Assistantship<br/>(IIT Delhi)</b> | <ul style="list-style-type: none"> <li>• Advanced Air Pollution Laboratory (CVP 820, Graduate course)</li> <li>• Contributed to course content development and lecture preparation</li> <li>• Assisted in class delivery and student engagement</li> <li>• Supported laboratory experiments and practical sessions</li> <li>• Assisted in preparing and grading assignments and examinations</li> </ul> |
| <b>2024</b> | <b>Teaching Assistantship<br/>(IIT Delhi)</b> | <ul style="list-style-type: none"> <li>• Environmental Science (CVL 100, Undergraduate Course)</li> <li>• Assisted in grading exams/assignments and contributed to preparation of question papers.</li> </ul>                                                                                                                                                                                           |